Supervised Protein-Language-Model Benchmark for TEM-1 Functionality Prediction

A classifier on ESM embeddings vs three bounding baselines

Benchmark Report

PREVIEW — benchmark tables not found on disk. This shows the report layout only; numbers populate after the Colab embeddings are dropped into data/features/plm_embeddings/ and notebooks 07/08 are re-run, then rebuild this PDF.

Task: predict functional vs non-functional (DMS_score_bin; positive = functional) for single missense TEM-1 beta-lactamase variants (Firnberg et al. 2014, BLAT_ECOLX) from **supervised classifiers trained on ESM embeddings**. **n = 4,783** variants across 263 residue positions; classes balanced (2,397 functional / 2,386 non-functional). This is arm-3: the learned-feature model the two no-training and no-language-model baselines exist to bound.

1. Why a supervised pLLM benchmark

The project's three prior arms set the bounds this model must beat (decision D027 keeps measured baselines, not asserted ones). **AA-identity (02)** is the no-language-model, raw-amino-acid floor: strong on the random split, collapsing on the contiguous region-holdout split because it can only memorize positions it trained on. **Physicochemical (04)** is the hand-engineered-descriptor control. **Zero-shot pLLM (06)** is the no-training control: a frozen ESM surprisal score, read off the model with no exposure to the labels. This notebook asks the arm-3 question directly: do **learned** embedding features beat all three — especially on the contiguous split — which would be the evidence that language-model representations carry signal beyond memorized positions and beyond the raw surprisal score.

The features are ESM embedding **deltas** between the mutant and wild-type sequence (decisions D035/D036): the change in the contextual embedding at the mutated residue (*delta_site*, the primary variant-specific feature), the whole-sequence mean delta (*delta_pooled*), and a local-window delta (*delta_local*). Unlike the single surprisal scalar of 06, these are high-dimensional vectors, which is exactly why a reduction guardrail is mandatory before modeling (below).

2. Methods

Features (D035/D036). Per variant, three embedding-delta blocks per ESM model, extracted on GPU by the companion Colab notebook 07a and read here from disk. The wild-type embedding is computed once per model and reused for all 4,783 deltas. The ESM ladder (D010–D013) spans ESM-1b, ESM-1v, ESM-2 (150M/650M/3B) and ESM-C (300M/600M).

Reduction guardrail (D037, non-negotiable). The raw feature matrix is ~10,000+ columns (three blocks × up to seven models). Feeding it whole into a tree swamps the model and, under the region-holdout splits, invites overfitting to the training positions. PCA (with standardization) is therefore fit **inside each cross-validation fold, on the training rows only**, per model, and applied to the held-out fold — never on the full data. No test row ever informs the scaler mean or the PCA basis. A leakage assertion confirms no seq_id / position / label-derived column enters the matrix and that no single dimension tracks the label almost perfectly.

Classifiers (D023–D025). L2 logistic regression, random forest, and XGBoost, plus a majority-class dummy floor. Per D026 no winner is pre-committed — the deployed model is selected empirically on the contiguous split (accuracy + calibration), the hardest and most realistic setting.

Splits, threshold, metrics. Identical to 02/04/06: five-fold random / modulo / contiguous splits (seed 42; the two position-based schemes are group-aware, no position in both train and test). The decision threshold is Youden's J fit on the **train fold only** and applied to the held-out fold. Metrics: ROC-AUC, PR-AUC, balanced accuracy, F1, MCC, and the domain-weighted utility (TP +1, TN +1, FN −0.25, FP −1); with bootstrap 95% CIs (2,000 resamples), DeLong and McNemar tests versus the best model per split, and the dummy floor.

3. Results

Table 1. Supervised-PLM cross-validated performance — populates from results/tables/08_pllm_supervised_benchmark_metrics_grid.csv once embeddings exist.

[metrics grid, model selection, and the four-way comparison populate here after the Colab embeddings are extracted and notebooks 07/08 re-run.]

4. Figures

[figure not found: four_way_comparison.pdf — run the notebook to generate]

[figure not found: roc_curves.pdf — run the notebook to generate]

[figure not found: metric_bars.pdf — run the notebook to generate]

> **What it shows.** The error bars are 2,000-resample bootstrap 95% intervals, so overlapping intervals mean the ranking between two learners is not statistically resolved (see the DeLong / McNemar tests in the significance table). Every learner clears the 0.50 floor on all three splits; the reliable comparison is each learner against the dummy and across splits, not small gaps between the top learners.

[figure not found: utility_bars.pdf — run the notebook to generate]

> **What it shows.** This reweights predictions by the project's real decision cost: a false 'functional' call (FP, −1) is four times as costly as missing a functional variant (FN, −0.25), because calling a dead resistance gene active is the dangerous error in a surveillance setting. The bar to watch is the contiguous split — a model whose utility holds there, not just on random, is the one that would survive deployment to unseen regions.

[figure not found: block_magnitude.pdf — run the notebook to generate]

> **What it shows.** This is the empirical justification for treating the site-level delta as the primary feature (D035). A single substitution barely moves the whole-sequence mean, so delta_pooled has a small per-variant norm; the change at the mutated residue itself (delta_site) is much larger and carries the variant-specific signal.

[figure not found: pca_scree.pdf — run the notebook to generate]

> **What it shows.** Each block is 640–2560-dimensional, but most of the variance lives in far fewer components. The curves' steep early rise is what makes the D037 reduction safe: projecting to ~50–200 components per model before the classifier discards width, not signal.

5. Interpretation and limitations

The result to read off Table 2 and Figure 1 is whether the supervised-PLM arm beats its three bounding baselines on the contiguous region-holdout split. Beating AA-identity there means the language-model representation, not position memorization, is doing the work; beating physicochemical means it exceeds hand-engineered chemistry; beating zero-shot means training on embeddings adds signal over the raw surprisal score. A supervised-PLM that only matches zero-shot on the hard split is itself a publishable finding: it says the scalar surprisal already captures most of the usable sequence-only signal for this protein, and the embedding complexity is not justified.

Guardrail (D037). The reported numbers are free of reduction-leakage: PCA is fit inside each fold on training rows only, no test row informs the projection, and no identity or label-derived column enters the feature matrix (asserted in the notebook). This is what makes the contiguous-split number trustworthy rather than an optimistic artifact of fitting the reducer on all data.

Limitation, stated where the number lives (D039). These embeddings, like the zero-shot scores, are sequence-only and shaped by foldability and stability, so they can under-detect a *catalytic-but-stable* knockout — an active-site substitution (Ambler S70, K73, S130, E166) that abolishes activity without destabilizing the fold. Training on embeddings tests whether the supervised head recovers signal the zero-shot score misses; it does not close the stability-vs-catalysis blind spot. That is the role of the ESMFold structural-epistasis features (D015/D038, for the double mutants) and the wet-lab mutagenesis panel, which check predictions against real selective-plating outcomes.